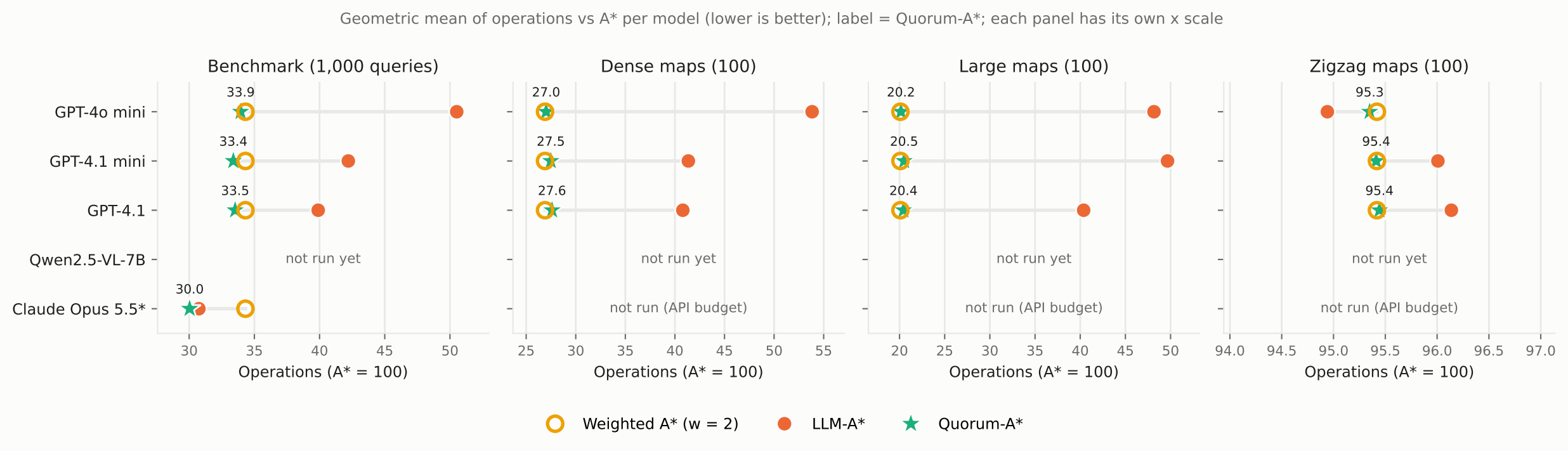
Geometric mean of operations vs A* per model (lower is better); label = Quorum-A*; each panel has its own x scale
Benchmark (1,000 queries)
Dense maps (100)
Large maps (100)
Zigzag maps (100)
GPT-4o mini
GPT-4.1 mini
GPT-4.1
Qwen2.5-VL-7B
Claude Opus 5.5*
33.9
33.4
33.5
30.0
27.0
27.5
27.6
20.2
20.5
20.4
95.3
95.4
95.4
not run yet
not run (API budget)
30
35
40
45
50
25
55
20
94.0
94.5
95.0
95.5
96.0
96.5
97.0
Operations (A* = 100)
Weighted A* (w = 2)
LLM-A*
Quorum-A*